MIL-C-10405/1A(EL)

18 September 1967

SUPERSEDING
MIL-C-10405/1(SigC)
2 April 1954

# MILITARY SPECIFICATION

## CRYSTAL UNIT USED IN TRANSMITTER BC-610( )

### 1. SCOPE

1.1 This specification covers the detail requirements of the crystal unit used with TRANSMITTER BC-610( ). These crystal units have a quartz vibrator pressure mounted in a plastic holder, type HC-22/U, and are designed to operate at anti-resonance into a load capacitance of 40 pf, on the fundamental frequency of the vibrator, under noncontrolled temperature conditions, at a specified frequency within the frequency range of 2.0 to 6.0 megahertz (MHz) , inclusive.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-10405 - Crystal Units, Quartz, Pressure and Spacer Mounted

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

MILITARY

SC-DL-12432 - Crystal Holder, HC-22/U, Assembly.
SC-B-22893 - Name Plate, Marking.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5955

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-C-10405 and as specified herein.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Sampling and inspection shall be in accordance with MIL-C-10405, and as specified herein.

Table I. Detail requirements.

| Requirement | Specified value or condition |
|---|---|
| Holder | HC-22/U (per drawing SC-DL-12432) |
| Dimensions | (See SC-DL-12432) |
| Frequency range | 2.0 to 6.0 MHz , incl. |
| Frequency tolerance | ±0.01 percent |
| Mode of oscillation | Fundamental |
| Load cap | 40 ±0.5 pf |
| Operating temperature range | -40°C to +70°C, incl. |
| Drive level, rated | (See Figure 1 of MIL-C-10405) |
| Equivalent resistance | (See Figure 1 of MIL-C-10405) |
| Government reference standard test set | TS-330/TSM |
| Springs | Minimum force, 2-1/2 lbs. |
| Marking | Per Drawing SC-B-22893 |
| Shock: | |
| Frequency change permitted | ±0.002 percent or 50cps, whichever is greater. |
| | The resistance shall not exceed those values shown in fig. 1 of MIL-C-10405 |
| Vibration[1]: | |
| Frequency change permitted | ±0.002 percent or 50cps, whichever is greater. |
| | The resistance shall not exceed those values shown in fig. 1 of MIL-C-10405 |
| Cleaning test | 250 cps (See 3.13 of MIL-C-10405) |

[1] Method 201 of MIL-STD-202, amplitude 0.06 inch total excursion.



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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-10405.

6. NOTES

6.1 The notes specified in MIL-C-10405 are applicable to this specification.

Custodian:
Army - EL

Preparing activity:
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